Venkata Lakshmi Parimala Pasupuleti

Principal Data Analyst · Financial Services Data Governance · Python · SQL · AWS · Statistical Modeling

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | McLean area, Virginia

PROFESSIONAL SUMMARY

Data Analyst with 8+ years delivering **data governance and quality management in financial services** — enforcing compliance standards, maintaining data lineage and metadata, and building audit-ready pipelines across 1M+ financial records at production scale. At Entrata, maintained **100% PHI compliance with zero violations** over 7+ years, tuned **200+ SQL queries for 30% performance improvement**, and built automated financial data pipelines on AWS (S3, EC2, RDS, Glue). Built multi-source statistical models achieving **$R^2 = 0.977$** across 190+ countries (Python, R) and **85%+ classification accuracy on 104K+ records**. Designed strategic BI dashboards improving decision-making comprehension **40%** (Tableau, ggplot2). Applied Isolation Forest anomaly detection across 3 pipelines, surfacing outliers that manual review missed. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

Entrata

Jan 2017 – Aug 2024

Analytic Software Engineer — Income Fraud Detection · Financial Data Governance · Compliance Analytics

- Reviewed and verified income documentation for **millions of tenant certifications** across Section 8/HCV, LIHTC, and HUD portfolios — identifying inconsistencies between stated income and supporting documents that flagged ineligible or misrepresented applications for escalation.
- Built **property performance dashboards** across 100K+ unit portfolios — eligibility approval/denial patterns, waitlist demand by bedroom size and income band, occupancy and vacancy rates, rent burden vs. AMI and voucher limits, delinquency rates, and lease renewal trends; applied LIHTC/Section 42 and HUD income-limit rules to validate unit mix, rent limits, and tenant documentation for compliance reporting across Section 8/HCV, HOME, and LIHTC programs.
- Eliminated **80% of manual eligibility review workload** with zero compliance violations by encoding the decision logic I had applied manually into automated Section 8/HCV, LIHTC, and HUD compliance pipelines — AMI threshold calculations, recertification workflows, and escalation routing across 100K+ unit portfolios.
- Achieved **100% PHI compliance** across all client data by migrating to AWS S3 with Libsodium-encrypted pipelines — zero security incidents; improved SQL query performance **30%** across 200+ queries supporting 1M+ property records at 99.5% uptime.

George Mason University

Aug 2024 – Aug 2025

Research Associate — Statistical Modeling · Strategic Analytics · Data Visualization

- Mined complex, multi-source datasets (WHO global health + UK Road Safety, 190+ countries) using open-source technologies (Python, R) — designed and trained regression models achieving **$R^2 = 0.977$**; end-to-end: data collection, feature engineering, cross-validation, pipeline automation.
- Surfaced statistically significant anomalies **manual review missed** across 3 research pipelines by applying Isolation Forest — translated findings into actionable research and program evaluation recommendations for faculty leadership.
- Partnered with faculty stakeholders to translate analytical outputs into strategic BI dashboards (Tableau, ggplot2, Quarto) — improved research comprehension and decision-making **40%** across two cohorts.

PROJECTS

CrashClarity: Strategic Analytics on 104K Records with Self-Service Dashboard

Python · R · Scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages — Live Dashboard

- Mined 104K+ UK accident records — built Random Forest + Logistic Regression ensemble with feature engineering achieving **85%+ accuracy**; deployed interactive self-service dashboard translating model outputs into strategic severity-reduction insights for non-technical stakeholders.

Leasel: Financial Services Data Quality & Compliance Audit Pipeline

Python · LangGraph · Scikit-learn · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built an end-to-end financial data quality and governance pipeline across 1,240 documents — feature extraction, rule-based classification, and deterministic audit engine with full metadata and lineage; achieved **93.6% precision, 87.9% F1** across 10 compliance rules; every determination traceable and audit-ready.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · Statistical modeling, data governance, ML pipelines, anomaly detection, data visualization

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017

TECHNICAL SKILLS

Data Governance & Quality: Data Quality Management, Metadata & Lineage, Audit Pipelines, PHI/HIPAA Compliance, Zero-Incident Data Migrations, LIHTC/HCV/HUD Regulatory Workflows, Access Governance, Security Controls

Analytics & Modeling: Python, R, SQL, PL/SQL, Statistical Modeling, Regression Analysis, Random Forest, Logistic Regression, XGBoost, Isolation Forest, Feature Engineering, Cross-Validation, Scikit-learn, Pandas, NumPy

Cloud & Data Infrastructure: AWS (S3, EC2, RDS, Glue), ETL Pipelines, PostgreSQL, MySQL, SQLite, Docker, CI/CD (GitHub Actions), Git, Linux

BI & Process: Tableau, Power BI, ggplot2, Quarto, matplotlib, Jupyter, Agile/Scrum (120+ sprints), FastAPI, REST APIs, JavaScript, TypeScript